

Lab 7: A Scaling Curve and Int8

Fit a toy power law, then quantize a weight matrix and measure error.

Work in a **Jupyter** notebook with **PyTorch** and `matplotlib`. Do notes **7.1–7.3** first. No Hugging Face weights this week.

First time you implement this. The lecture notes this week already answered what the method is, why it exists, the architecture, the formula, and the tradeoffs. Read that first-time block before these exercises. This lab is the first time you **compute** it, not the first definition.

```
import torch
import matplotlib.pyplot as plt
```

1. Fit a toy scaling law

Construct compute C and loss $L = aC^{-b} + c$ plus a little noise. You will recover a, b, c .

```
torch.manual_seed(0)
a_true, b_true, c_true = 2.0, 0.15, 1.4
C = torch.logspace(2, 6, 16)          # 1e2 .. 1e6
L = a_true * C.pow(-b_true) + c_true
L = L + 0.02 * torch.randn_like(L)
```

1. Scatter $\log C$ versus L . Guess whether c is near 1.4 before you fit.
2. Fit with Adam on unconstrained parameters, but keep $a > 0$, $b > 0$ via softplus:

```
p = torch.nn.Parameter(torch.tensor([1.0, 0.1, 1.0])) # raw a,b,c
opt = torch.optim.Adam([p], lr=0.05)
for _ in range(2000):
    a, b, c = torch.nn.functional.softplus(p[0]), torch.nn.functional.softplus(p[1]), p[2]
    pred = a * C.pow(-b) + c
    loss = ((pred - L) ** 2).mean()
    opt.zero_grad(); loss.backward(); opt.step()
```

Print fitted a, b, c versus truth. Overlay `pred` on the scatter.

3. In one sentence: if you only observed the four smallest C , would c still be identified?

2. Affine int8-ish quantization

```
torch.manual_seed(1)
x = torch.randn(256)
qmin, qmax = -128, 127
```

1. Compute `scale = (x.max() - x.min()) / (qmax - qmin)` and `zp = qmin - x.min() / scale`. Quantize with

```
q = torch.clamp(torch.round(x / scale + zp), qmin, qmax)
xhat = (q - zp) * scale
```

Print MSE `(x - xhat).pow(2).mean()`. It should be small relative to `x.var()`.

2. Repeat after replacing `x` with `x.clone(); x[0] = 50.0` (one outlier). MSE should jump. Why do activation outliers break naive INT8?
 3. Histogram `x` and `xhat` on one figure.
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3. Write-up

Four sentences: what C^{-b} says about diminishing returns, why Chinchilla would not only grow a 's "model" without growing tokens, what scale and zero-point are for, and one reason speculative decoding is not the same as distillation.